

MOHAMED TAREK SALAH

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OBJECTIVE

- Software Engineer that excel in adapting to new challenges and environments, continuously seeking opportunities to learn and apply my skills effectively to drive success in diverse projects.

EDUCATION

B.S. in Software Engineering, Helwan University in Cairo

Sep 2021 – Jun 2025

- GPA: **3.87**.

Graduation project (Grade: A+)

- Intelligent Sports Recommendation System for Children built using a rule-based engine (Drools) and image analysis techniques. The system allows parents to assign their children a series of physical tests (e.g., Standing Long Jump Test, One-Leg Stand, Sit-up Test, 30-Meter Sprint , ...etc), then provides personalized sports category suggestions, improvement tips, and training guidance based on performance. It also includes symmetry analysis using front and side child images to assess physical balance. The admin dashboard enables full management of sports, tests, and category data.

EXPERIENCE

Backend Developer

ISTQSERVER

Oct 2024 – Present

- Enhanced and scaled backend architecture across **4 production projects**, improving performance and scalability in over 75% of them.
- Designed and implemented a reusable **filtering class**, applicable across multiple projects, reducing code duplication and ensuring consistency in data handling.
- Collaborated with IT and network teams to deliver real-world functionality for multi-location deployments (e.g., **Madrid and Valencia**).
- Assisted with **production infrastructure and real-time features**, including configuring and updating **Nginx reverse proxy** settings to support persistent WebSocket connections and enable real-time communication between clients and production servers.
- Developed and maintained **comprehensive API documentation** with Postman, covering 100% of backend endpoints.

SKILLS

Programming Languages: C, C++, Java, JavaScript, TypeScript, Python

Testing: Manual Testing, API Testing (Swagger, Postman), E2E Test Automation (Jest)

Cloud: Cloudinary, Docker

Development: ReactJs, Node.js(Nest.js, Express.js) , Databases (PostgreSQL(TypeOrm), MongoDB)

Additional: OOP, Design Patterns, Git/GitHub, Software Development Lifecycle , Data Structures

PROJECTS

I have developed numerous projects showcasing a variety of skills and frameworks. GitHub profile: <https://github.com/Tar2sh12>.

E-Commerce Back-End Development Project

[GitHub](#)

- Re-engineered a scalable e-commerce backend using **NestJS, Node.js, MongoDB, and Mongoose**, applying modular architecture, Repository Pattern, and optimized database queries with indexing.
- Implemented secure **JWT authentication and role-based authorization** with access/refresh tokens, OTP verification, password reset, token blacklisting, password hashing, and sensitive data encryption.
- Developed complete e-commerce workflows including **products, cart, orders, reviews, coupons, and addresses**, with **Stripe Checkout and webhook-based payment verification**.
- Integrated **Socket.io real-time notifications**, Cloudinary image management, and NestJS middleware/interceptors for validation, logging, standardized responses, and performance monitoring.

Cinema Reservation System

[GitHub](#)

- Developed a Cinema Reservation System using **React.js, Express.js, MongoDB, and Redis**, with real-time admin notifications via Socket.io for seat reservations on specific dates.

Real-time Restaurant Reservation System

[GitHub](#)

- A real-time restaurant reservation system that lets users book a table at their favorite restaurant.
- Technologies: **Node.js, Express.js, MongoDB, MQTT, Angular**.

EXTRACURRICULAR ACTIVITIES

Problem Solving Instructor, Microsoft Student Partner

Aug 2022 – Sep 2022

- Instructed peers on C++ fundamentals, focusing on best practices and coding standards.
- Conducted workshops to help students succeed in competitive programming on platforms like Codeforces and LeetCode.

Object-Oriented Programming Member, Microsoft Student Partner

Sep 2023 – Oct 2023

- Taught OOP with C++ and guided a booking system project using core OOP principles.